

Agentic Prescriptive Analytics for Banking, from Prediction to Action

Turning Existing Predictions into Governed, Optimized, Recommended Actions

1. Background

Banks and credit unions already run predictive analytics. They score who is likely to default, who might leave, which loans could slip into non-performing, and which transactions look like fraud. This is mature work. Most institutions have done it for years.

The failure is what happens after the model runs. A number lands on a dashboard, and a human still has to work out what to do about it. Usually there is no time, no consistent method, and no memory of what worked last time. The prediction gets made. The decision often does not, or it gets made differently by every person who touches it.

Deciding what to do about a prediction is prescriptive analytics. It was historically expensive and complex, so only the largest banks, the ones with deep data science and operations benches, could afford it. Community banks and credit unions were priced out.

Agentic AI changes that math. It does not replace the models. It collapses the cost of everything around the decision: pulling the data, applying the rules, explaining the recommendation, drafting the action, and standing up the interface. Prescriptive analytics is now within reach for the smaller institutions that could never staff it.

We call the capability that closes this gap **Agentic Prescriptive Analytics**: an AI system that sits on top of an institution's existing predictions and data, estimates the effect of each candidate action, simulates and optimizes the resulting policy, and turns the result into clear, governed, recommended actions, each following the institution's own policies and routed through a human for approval.

1.1 What Prescriptive Analytics Is, and How It Differs From Prediction

A **predictive** model answers a question about the world: how likely is this account to default? It produces a probability or a forecast. It says nothing about what to do.

A **prescriptive** system answers a question about a decision: given everything we could do and everything that limits us, what should we actually do? A prescription needs three things a prediction does not supply on its own: an **objective** (what are we maximizing), **levers** (what actions are available), and **constraints** (what limits us). Sorting accounts by their risk score and working the list top down is not prescriptive analytics, even if an AI agent produces the sorted list, writes the call scripts, and explains the ranking in fluent prose. It is a predictive score with a narration layer on top. Two things expose why a sorted list is the wrong answer, not just a simplified one.

- **Risk is not value.** The account most likely to default is not necessarily the account worth calling first. A borrower who owes \$300 with no assets is a near-certain loss but a small one. A borrower who owes \$45,000 with a moderate default probability is a much larger expected loss and often more recoverable. Ranking on probability alone routinely prioritizes the wrong accounts.
- **Risk is not influenceability.** Some accounts will pay regardless of whether anyone calls them. Some will default regardless of what anyone offers. Spending capacity on either group wastes it. The accounts worth working are the ones where an action actually changes the outcome, and a risk score cannot tell you that, because it was never trained to.

A prescriptive system has to close both gaps: value the accounts in dollars, not probability, and target the ones an action can actually move. That is a different modeling and computation problem from producing the original risk score, and it is the part of the challenge that separates a real solution from a sorted list with a rationale bolted on.

1.2 How Prescriptive Analytics Is Actually Done

In practice, prescriptive analytics is built from four techniques used together, each answering a question the risk score cannot.

- **Expected-value framing.** Convert probability into a dollar figure so accounts are compared on what is actually at stake, not on a raw likelihood. A common form is expected loss, probability of default multiplied by exposure and by loss-given-default, net of the cost of acting. This alone changes the ranking, because a lower-probability, higher-balance account can outrank a higher-probability, low-balance one.
- **Causal or uplift modeling.** Estimate the effect of a specific action on a specific account, not just the account's risk without it. This is a genuinely different model from the risk model: it is trained to separate accounts into those who will pay regardless (a sure thing), those who will default regardless (a lost cause), those an action can win over (a persuadable), and, in some domains, those an action can backfire on (a sleeping dog). Techniques include T-learners, S-learners, X-learners, and causal forests; EconML and CausalML are the common libraries. Only the persuadables justify spending scarce capacity on.
- **Simulation.** Before committing a day's worklist to real customers, simulate it. Monte Carlo simulation samples across the uncertainty in the risk score, the uplift estimate, and payment behavior, and produces a distribution of likely outcomes, recovered dollars, promises to pay, staff hours used, rather than a single point guess. Discrete-event simulation (commonly built with SimPy) models the collections floor itself as a queue with arrivals and staff service times, which is how capacity and throughput questions get answered honestly rather than assumed. This is what a genuine what-if comparison looks like: policy A against policy B, compared on their outcome distributions, before either touches a real borrower.
- **Constrained optimization.** The daily decision is an assignment problem: choose a treatment for each account, a call, an SMS, a payment plan, a hardship offer, or no action, so as to maximize total expected, uplift-adjusted recovery, subject to real constraints: total staff call-minutes, contact-frequency and cooling-off limits, hardship eligibility, channel limits, budget. This is a mixed-integer program, solved with a real solver, OR-Tools, CBC, HiGHS, or commercially Gurobi or CPLEX, typically modeled through Pyomo or PuLP. The result is provably different from a sorted list under a binding capacity constraint: the solver will often skip an expensive, high-risk account in favor of two cheaper, higher-combined-uplift accounts, because that is what actually maximizes recovery. That

gap between the optimizer's choice and a plain sort is the clearest evidence that real prescriptive work happened.

Collections is not really a one-shot decision, it is sequential, since today's action changes tomorrow's right action. Framed formally, that is a **Markov Decision Process**, solvable by dynamic programming or reinforcement learning. It is a reach target for this hackfest, not a requirement.

Put together, this is the pipeline **Agentic Prescriptive Analytics** runs on top of an existing prediction:

Prediction → Causal Estimate of Action Effect → Simulated Policy Comparison → Optimized Assignment (Solver) → Policy-Aware Recommendation → Plain-Language Rationale → Human Approval → Action → Recorded Outcome

The agent's job in this pipeline is orchestration and explanation: call the causal model, call the simulator, call the solver, read back their outputs, check them against policy, draft the outreach, and write the rationale in plain language. The agent should not be the one estimating an uplift, running a simulation, or solving an assignment problem in its own reasoning. That division of labor is the core technical bet of this challenge.

2. The Opportunity

Imagine a system that does not stop at the prediction, and does not stop at a sorted list either. It estimates what each action would actually do, simulates the resulting policy before anyone commits to it, optimizes the day's assignment under real constraints, explains the result, drafts the action, and puts it in front of a human to approve.

Instead of leaving a collections manager to read a risk score off a dashboard and improvise, Agentic Prescriptive Analytics reads the existing predictions, estimates which accounts an action can actually influence, simulates candidate policies, optimizes the day's assignment under capacity and hardship constraints, recommends a specific action per account, drafts the outreach, and attaches a plain-language reason. A person reviews and approves before anything goes out.

A collections manager could ask: “who do I contact today, in what order, what do I offer, and why?” and receive a ranked, optimized list, a recommended action per account, drafted outreach, a reason for each call grounded in both the account's value and its estimated influenceability, and an approval step before any contact is made. The recommendations respect the institution's hardship policies and the team's capacity.

Agentic Prescriptive Analytics is therefore not a new predictive model, a dashboard, a sorted list with commentary, or an autonomous system that acts on its own. It does not do the risk scoring itself. Its value lies in closing the gap between the prediction and the action, with real causal, simulation, and optimization work underneath, affordably and with governance, so that a prediction becomes a decided, approved, and executed step. It connects:

Prediction → Causal Estimate of Action Effect → Simulated Policy Comparison → Optimized Assignment → Policy-Aware Recommendation → Plain-Language Rationale → Human Approval → Action → Recorded Outcome

For a services firm, this brings genuine prescriptive analytics, not a relabeled sort, within reach of the community banks and credit unions that were priced out of it, and does so on top of the predictions they already run.

3. Challenge Statement

Your challenge is to design and build (prototype) an AI system that sits on top of a bank's existing predictions and data, estimates the effect of candidate actions, simulates and optimizes the resulting policy, and turns the result into clear, recommended actions: actions that follow the institution's own policies, carry a plain-language rationale, and route through a human for review and approval. In one line, close the gap between “here is the prediction” and “here is what we did about it,” and do it with real causal and optimization work, not a sorted score.

To keep it concrete, teams work one real scenario: collections and delinquency, end to end. A collections manager should be able to ask who to contact today, in what order, what to offer, and why, and get back a ranked and

optimized list, a recommended action per account, drafted outreach, a reason for each call, and an approval step before anything goes out. The system respects the bank's hardship policies and its staff's capacity.

Collections is a deliberate choice. It is high-value and lower-risk, an operational task rather than a lending decision, so it sits clear of the heavier loan-approval regulations.

Past getting the scenario running, teams are expected to show four things:

- **The right division of labor with AI.** The AI orchestrates and explains. It hands the actual math, the causal effect estimate, the policy simulation, and the constrained optimization, to proper models and solvers instead of reasoning its way through the numbers itself. How cleanly a team draws that line is a core judging point. It is the difference between a solution that holds up and a demo that collapses the moment the numbers matter.
- **Governance built in from the start.** Every recommendation carries a reason, an audit trail, and a human approval step, the kind that survives a bank's compliance and risk review. In this industry that is not a feature. It is what makes the thing sellable at all.
- **A design that reaches past collections.** Build the collections slice, but architect the engine so the same foundation later takes NPA early warning, member retention, or fraud triage by swapping in a new scenario, a new causal target, and new constraints. Ship the slice; design the reusable core.
- **A client-ready story.** As a services company, teams also carry the business case: who the client is, what it saves them, and how we take it to market.

The objective is not to build the full multi-scenario platform during the hackfest. Teams should prototype the collections scenario end to end, with a real causal, simulation, and optimization layer underneath, while designing the reusable core beneath it. Teams are free to determine the AI models, agent framework, causal method, simulation approach, solver, policy encoding, and supporting technologies required.

4. Scope

The objective of this challenge is not to build another predictive model or another dashboard, and it is not to build a sorted list with an AI narration layer on top. Teams are expected to turn predictions that already exist into governed, approved actions, with the causal estimation, simulation, and optimization done by proper models and solvers.

In Scope

The proposed solution should consider capabilities such as:

- Ingesting existing predictions, such as delinquency and default risk scores, together with the account data around them.
- Framing the objective in expected value, for example probability of default multiplied by exposure and loss-given-default, net of the cost of acting, rather than ranking on raw probability.
- Estimating the causal effect, the uplift, of each candidate action on each account, so that accounts are prioritized by how much an action can change the outcome, not only by how risky the account is.
- Simulating candidate policies, using Monte Carlo and/or discrete-event methods, to compare their range of likely outcomes, recovery, promises to pay, staff hours, before any policy is committed.
- Solving a constrained optimization to assign a specific action per account for the day, using a real solver, not the LLM, subject to hardship policy, contact-frequency limits, and staff capacity.
- Encoding hardship and collections policy as machine-checkable constraints rather than model guesswork.
- Respecting the team's real capacity as a constraint on the daily worklist, ideally reflected in the simulation and the optimizer, not just capped after the fact.
- Drafting the outreach, such as a call script or message, for each recommended action.
- Attaching a plain-language rationale to every recommendation, one that reflects both the account's value and its estimated influenceability, and, where relevant, why the optimizer chose it over another account.

- An audit trail and a human review-and-approval step before any action goes out.
- Recording outcomes so the causal estimate and the simulator can be recalibrated over time, giving the system a memory of what actually worked.
- Architecting the orchestration engine so a new scenario, such as NPA early warning, retention, or fraud triage, plugs in by swapping the scenario, the causal target, and the constraints, rather than rebuilding the foundation.

Teams are expected to explain how these capabilities collectively deliver Agentic Prescriptive Analytics. The prototype should demonstrate the collections scenario end to end.

Out of Scope

The following are outside the scope of this challenge:

- Building new predictive risk models. Assume the delinquency and default predictions already exist. (The causal or uplift model, which estimates the effect of an action, is a different model from the risk model and is in scope.)
- Letting the LLM perform the risk scoring, the causal estimation, the simulation, or the optimization itself, or approximate any of them through reasoning alone.
- Acting autonomously without human review and approval.
- Lending or credit-approval decisions. Collections is operational and is deliberately chosen to sit clear of loan-approval regulations.
- Building a data platform or replacing the core banking system.
- A generic chatbot over bank data.
- A full sequential, multi-period optimization (a full Markov Decision Process formulation) across the life of a delinquency. This is a valid reach target for stronger teams, not a requirement.

The emphasis should remain on turning existing predictions into governed, approved action, with orchestration and explanation from the AI and the causal, simulation, and optimization math from proper models and solvers.

5. Reference Scenario: Collections and Delinquency

For this challenge the scenario is fixed. Teams work collections and delinquency end to end.

Assume you are working with a community bank or credit union that already scores delinquency and default risk. The scores land on a dashboard. A collections manager and a small team work the accounts each day with limited time, no consistent method, and no memory of what worked last time. Hardship policies exist but are applied inconsistently. Capacity is tight, and the team can only make so many contacts in a day.

The manager wants to ask a single question: who do I contact today, in what order, what do I offer, and why. The system should return a ranked and optimized list, a recommended action per account, drafted outreach, a reason for each call, and an approval step before any contact is made. Every recommendation should respect the institution's hardship policy and the team's capacity.

Collections is chosen deliberately. It is high-value and lower-risk, an operational task rather than a lending decision, so it sits clear of the heavier loan-approval regulations. The engine built for it, however, should generalize. A later scenario such as NPA early warning, member retention, or fraud triage should reuse the same foundation, with its own causal target and its own constraints.

Note on the synthetic data. Where organizers provide a synthetic collections portfolio, the true effect of each action on each account type may be withheld from teams by design, the way it would be unknown in a real portfolio. Teams should treat their causal and optimization layer as an estimate of that unknown effect, not as access to it. This is what makes the exercise honest: a plain sort will visibly leave recoverable value on the table against a policy that correctly targets influenceable accounts.

6. Representative Business Questions

Agentic Prescriptive Analytics should demonstrate its ability to move from a prediction to a governed, optimized action, not just surface a score or a sorted list.

The Daily Decision

- Who should we contact today, and in what order?
- What action should we take on this account, and what should we offer?
- Why this account, and why this action?
- What outreach goes out, and who approves it before it does?

Value and Influence

- What is this account actually worth in expected dollars, not just its risk probability?
- Which accounts are persuadable, a sure thing, or a lost cause, and how does that change today's worklist?
- What is the estimated effect of this specific action on this specific account?

Simulation and What-If

- What would today's policy recover on average, and how much does that vary?
- What would a different policy have recovered, before we commit to it?
- Does today's worklist fit the team's capacity, and where does it break if volume spikes?

Policy and Optimization

- Which hardship options is this borrower eligible for under our policy?
- Why did the optimizer choose this account over that one, given the same capacity?
- Does this recommended action comply with our collections and hardship policy?

Division of Labor

- Which model, simulation, or solver produced this ranking, this uplift estimate, and this assignment?
- What did the AI decide, and what did it hand to a model, a simulator, or a solver?
- Where in this recommendation is a number the LLM should not have computed itself?

Governance and Audit

- What is the rationale for this recommendation, in plain language?
- Who reviewed and approved this action, and when?
- What is the audit trail from prediction to approved action?

Outcome and Reuse

- What happened after we took this action last time, and what worked for similar accounts?
- How would NPA early warning, retention, or fraud triage reuse this engine?
- What has to change to add a new scenario, and what stays the same?

These questions are illustrative. Teams are encouraged to identify additional scenarios that demonstrate governed, optimized action rather than a bare prediction or a sorted list.

7. Design Principles

Teams are free to determine the detailed architecture and implementation approach, but the proposed solution should adhere to the following principles.

- **Causal, Not Just Predictive.** The system estimates what an action changes, not only how risky an account is. Prioritization reflects influenceability, not just probability of default.
- **Valued in Expected Dollars.** Accounts and actions are compared on expected value, risk times exposure times severity, net of cost, not on raw probability.

- **Simulated Before Committed.** Candidate policies are compared over a distribution of likely outcomes through simulation before a worklist is finalized, not assumed to work from a single point estimate.
- **Models and Solvers Do the Math.** The AI orchestrates and explains. Causal effect estimation, policy simulation, and assignment optimization go to proper models, simulators, and solvers. The LLM should not reason its way through the numbers. How cleanly this line is drawn separates a solution that holds up from a demo that collapses the moment the numbers matter.
- **Recommend, Do Not Act.** Every recommendation routes through a human for review and approval before anything goes out. The system proposes; a person disposes.
- **Governed by Construction.** Every recommendation carries a reason, an audit trail, and an approval step that survives a bank's compliance and risk review. In this industry governance is not a feature. It is what makes the solution sellable at all.
- **Policy-Aware.** Recommendations follow the institution's own hardship and collections policies, encoded as constraints, rather than the model's guess at good practice.
- **Capacity-Aware.** The daily worklist respects the team's real capacity, reflected as a constraint in the optimization and, ideally, in the simulation. A recommendation the team cannot execute is not useful.
- **Explainable in Plain Language.** Every recommendation carries a rationale that a collections manager and a compliance reviewer can both read, including why an account was or was not selected.
- **Ship the Slice, Design the Core.** Build collections end to end, but architect the engine so a new scenario plugs in by swapping the scenario, the causal target, and the constraints, not rebuilding the foundation.
- **Has Memory.** The system records outcomes so it can recalibrate its causal estimates and its simulator over time, rather than starting cold every day.
- **Client-Ready.** As a services firm, the solution must carry a business case: who the client is, what it saves, and how it goes to market.

8. Expected Deliverables

Each team is expected to present a complete solution vision supported by a working demonstration.

- **Working Demonstration.** The collections scenario end to end, from an existing prediction through causal estimation, simulation, and optimization to a recommended action and a human approval.
- **Solution Architecture.** The orchestration engine, showing where the AI orchestrates and explains and where the causal model, the simulator, and the solver do the math, together with the governance and approval path.
- **Causal and Optimization Layer.** The uplift or causal method used, the simulation approach, and the solver and constraint formulation used to produce the daily assignment, and why each choice fits the problem.
- **Division of Labor.** An explicit account of what the AI decides and what it hands to models, simulators, and solvers, and why the line sits where it does.
- **Governance Model.** How every recommendation carries a reason, an audit trail, and a human approval step, and how this survives a bank's compliance and risk review.
- **Reusable Core.** How the same engine takes a new scenario, such as NPA early warning, retention, or fraud triage, by swapping the scenario, the causal target, and the constraints, rather than rebuilding.
- **Business Case.** Who the client is, what the solution saves them, and how a services firm takes it to market.
- **Future Roadmap.** A brief explanation of how the prototype could evolve into an enterprise-grade Agentic Prescriptive Analytics capability, including the sequential, multi-period extension for stronger teams that reach for it.

Every submission should demonstrate the ability to:

- **Value.** Frame accounts and actions in expected dollars, not raw probability.
- **Target.** Estimate which accounts an action can actually influence, using a causal or uplift method, not the risk score alone.

- **Simulate.** Compare candidate policies over a distribution of likely outcomes before committing to one.
- **Optimize.** Solve the daily assignment as a constrained optimization, using a real solver, not the LLM.
- **Recommend.** Propose a specific, policy-compliant action for each account.
- **Explain.** Attach a plain-language rationale to every recommendation.
- **Govern.** Carry an audit trail and route through human approval before any action.
- **Reuse.** Take a new scenario by swapping the scenario, the causal target, and the constraints, not rebuilding the engine.

9. Evaluation Criteria

Criteria	Weightage
Business Value	30%
Technical Innovation	30%
Solution Architecture	30%
Presentation & Demonstration	10%

Business Value (30%)

- Assesses how well the solution addresses a meaningful business problem.
- Measures relevance to business or customer needs.
- Considers potential to improve efficiency, outcomes, or decision-making.
- Evaluates applicability to real-world enterprise scenarios.

Innovation (30%)

- Assesses the originality and creativity of the solution.

- Evaluates meaningful use of AI beyond basic or surface-level capabilities.
- Considers technical creativity in automation, reasoning, and orchestration.
- Measures how effectively technology is applied to solve the identified problem.

Solution Architecture (30%)

- Assesses the quality, feasibility, and scalability of the solution architecture.
- Evaluates component design, integration, and extensibility.
- Considers enterprise readiness, including security and governance.
- Reviews Responsible AI, explainability, auditability, traceability, and human-in-the-loop considerations where applicable.

Presentation & Demonstration (10%)

- Assesses clarity in communicating the problem, approach, and solution.
- Evaluates the quality and flow of the prototype demonstration.
- Considers use of realistic scenarios to explain solution impact.
- Measures how well business value, technical value, and responses to questions are articulated.